

Saqib Mehdi

Curriculum Vitae

Institute of Geodesy and Geoinformatics, UPWr, Poland

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Education

- 2022–2026 **PhD, Civil Engineering, Geodesy & Transport, Uniwersytet Przyrodniczy we Wrocławiu (UPWr), Poland**
Research focused on urban tropospheric sensing using GNSS on moving platforms, such as trams, to estimate Zenith Tropospheric Delay. Developed methods integrating urban ray-tracing, multipath mitigation, and GNSS Precise Point Positioning to enable high-resolution atmospheric observations in metropolitan environments. Applications include improving urban-scale numerical weather models, enhancing weather forecast accuracy, and assessing optimal observation zones in dense city areas.
- 2018–2022 **Master of Science, Global Navigation Satellite System [link](#), Institute of Space Technology, Pakistan**
Specialized in satellite navigation systems, GNSS signal processing, and geospatial data analysis. Master's thesis focused on detecting seismo-tropospheric perturbations over earthquake-prone regions. Conducted analysis of GNSS-derived tropospheric delays and Total Electron Content (TEC), alongside atmospheric gases (SO₂ and SO₄), to identify anomalies preceding and following seismic events, demonstrating the coupling between lithosphere and atmosphere for earthquake monitoring.
- CGPA : 3.8/4
- 2014–2018 **Bachelor of Science, Physics [link](#), University of the Punjab, Pakistan**
Major subjects included Classical Electrodynamics, Advanced Electronics, Solid State Physics, and Fundamental & Digital Electronics. Coursework also covered quantum mechanics, thermal physics, mathematical methods, differential equations, and computational techniques in C++ and MATLAB.
- CGPA : 3.21/4

Publications

Journal Articles

- 2026 **Saqib Mehdi, Mingda Ye, Guohao Zhang, Witold Rohm**, Ray-Tracing-Based Multipath Classification and Adaptive Satellite Selection for Improved Tropospheric Estimation, , *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTAR)* (<https://doi.org/10.1109/JSTARS.2026.3697547>)
Accurate estimation of Zenith Tropospheric Delay (ZTD) in dense urban environments is challenged by multipath and NLOS signals. This study introduces a ray-tracing-assisted PPP framework that models signal reflections and diffractions, adaptively reweighting or excluding contaminated signals. Results show sub-meter positioning and significant improvement in ZTD accuracy.
- 2026 **Saqib Mehdi, Marcus Franz Wareyka-Glaner, Guohao Zhang, Witold Rohm**, Identifying healthy urban zones for GNSS-based tropospheric delay estimation using 3D ray-tracing, *GPS Solutions* (<https://doi.org/10.1007/s10291-026-02065-1e>)
Presents a method to identify urban locations ("healthy zones") optimal for GNSS-based ZTD estimation using 3D building models and ray-tracing. Locations with at least four clean LOS satellites are selected, improving the robustness of crowdsourced tropospheric observations for high-resolution weather forecasting.

- in progress **Saqib Mehdi, Witold Rohm**, Ray-Tracing-Based Multipath Classification and Adaptive Satellite Selection for Moving GNSS Receivers, *IEEE JSTAR* (Work in Progress)
Focuses on mitigating urban multipath effects and improving the accuracy of GNSS estimates for a moving receiver in metropolitan environments.
- 2021 **Saqib Mehdi, Munwar Shah, Najam Abbas Naqvi**, Lithosphere–Atmosphere–Ionosphere Coupling Associated with the 2019 Mw 7.1 California Earthquake Using GNSS and Multiple Satellites, *Environmental Monitoring and Assessment*, 193, 501 (2021).
Investigated lithosphere–atmosphere–ionosphere coupling effects associated with the 2019 Ridgecrest (Mw 7.1) California earthquake using GNSS and multi-satellite observations. Analyzed ionospheric perturbations and atmospheric responses to understand pre- and post-seismic coupling mechanisms. (<https://doi.org/10.1007/s10661-021-09278-6>)

Conference Presentations

- 2025 **Poster Presentation** [link](#), Ray-Tracing for Precise GNSS Troposphere Estimation in Urban Environments, European Navigation Conference (ENC)
- 2026 **Oral Presentation**, Ray-Tracing–Assisted Urban GNSS for Robust Zenith Tropospheric Delay Estimation, ENC ([Presentation Link](#))
- 2026 **Oral Presentation**, Geometry-Aware PPP for Reliable GNSS Tropospheric Sensing in Dense Urban Environment, European Geosciences Union (EGU) (<https://doi.org/10.5194/egusphere-egu26-18365>)

Research Visits

- January – February, **GNSS Research Intern** [link](#)
- 2022 Worked in the Ground-Based GNSS Augmentation Group on a feasibility study for Continuous Operating Reference Stations (CORS) network deployment in Islamabad. Contributed to analysis and planning of GNSS augmentation infrastructure for improved positioning accuracy.
Advisor : **Dr. Najam Abbas Naqvi**, *Institute of Space Technology, Pakistan*, ([Personal Web-page](#))
- April, 2024 **Training on Spirent GNSS Simulator**
Received specialized training on Spirent GNSS hardware simulator for generating and analyzing GNSS signals. Gained hands-on experience in tropospheric data simulation and PPP processing in controlled environments, supporting urban GNSS troposphere research.
Advisor : **Dr inż. Marcin Szołucha**, *Military University of Warsaw, Poland*, ([Personal Web-page](#))
- August, 2024 **Research on GNSS Tropospheric Estimation Using Mobile Receivers in Urban Environments** [link](#)
Conducted research on GNSS tropospheric estimation using mobile GNSS receivers in urban environments. Utilized the Spirent hardware simulator to generate and analyze simulated GNSS signals, processing data to derive tropospheric estimates.
Advisor : **Prof. Thomas Hobiger**, *Institute of Navigation, University of Stuttgart, Germany*, ([Personal Web-page](#))
- July – August, **Urban GNSS Troposphere Estimation and Ray-Tracing Simulation Techniques** [link](#)
- 2025 Engaged in research on GNSS data processing, urban troposphere estimation, and ray-tracing simulation techniques to study multipath effects. Contributed to the integration of urban ray-tracing with PPP algorithms for Zenith Wet Delay (ZWD) retrieval and developed classifier-based methods for multipath detection and exclusion.
Advisor : **Prof. Guohao ZHANG**, *Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University*, ([Personal Web-page](#))

September, 2025 ***Techniques for GNSS-based Positioning and Atmospheric Sensing*** [link](#)
Focused on advanced filtering techniques in GNSS-based positioning with low-cost receivers. Applied Kalman filtering for Precise Point Positioning (PPP), noise reduction, and multipath mitigation in urban environments. Integrated filtering and multipath strategies to improve robustness of GNSS-based atmospheric sensing.

Advisor : **Dr. Hab. Eng. Jacek Paziewski**, *Faculty of Geoengineering, University of Warmia and Mazury, Olsztyn, Poland*, ([Personal Web-page](#))

Summer Schools & Trainings

2016 –present ***ESA-JCR GNSS Summer School, Slovenia*** [link](#) Participated in a comprehensive 10-day program on satellite navigation, including Differential and Precise Point Positioning (PPP), GNSS signal processing, ionospheric and tropospheric effects, and multi-frequency/multi-system GNSS (Galileo, GPS, etc.). Gained hands-on experience in GNSS measurement generation, challenging environment navigation, cyber-security, and satellite-based augmentation systems. Developed teamwork, leadership, and GNSS entrepreneurship skills through lab exercises and project-based learning.

2019 ***Introduction to Mathematical and Computational Modelling for Sustainability, Oradea University, Romania*** [link](#) Participated in a blended intensive Erasmus+ program focused on mathematical and computational modeling for sustainability. Learned techniques for modeling environmental systems and sustainability challenges, combining theoretical foundations with practical, team-based problem solving.

Teaching Assistantship

Atmospheric Monitoring using GNSS Polarimetric Radio Occultation, UPWr, Poland
Introduction to MATLAB Programming, UPWr, Poland

Computer Skills

Programming Python (NumPy, SciPy, Pandas), MATLAB, C++
Document $\text{\LaTeX}$
Preparation

Extra-Curricular Activities

2024 **Representative for Foreign Students Affairs** [link](#), PhD Student Council, UPWr, Poland.
to represent international PhD students and support academic internationalization initiatives.

References

Prof. Witold Rohm [link](#)
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Dr. Guohao Zhang [link](#)
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Dr. Saeid Haji-aghajany [link](#)
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